Points to Remember

- Probability of an event:

  (i) Classical Method: $P(A) = \dfrac{No.\ of\ outcomes\ in\ favour\ of\ A}{No.\ if\ elements\ in\ samplespace}$

  (ii) Relative Frequency Method: $P(A) = \underset{n \to \infty}{Lim} \dfrac{n(A)}{n}$

- Axioms

  (i) $P(A) \geq 0$

  (ii) $P(S) = 1$

  (iii) If $\{A_1, A_2, \ldots\}$ is a sequence of mutually exclusive events, then

  $P(A_1 \cup A_2 \cup \ldots) = P(A_1) + P(A_2) + \ldots$

- Probability of complementary event: $P(\bar{A}) = 1 - P(A)$

- Addition theorem of probability: $P(A \cup B) = P(A) + P(B)$, if A and B are mutually exclusive events

  $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

- If $A \subset B$, then $P(A) \leq P(B)$ and $P(B\text{-}A) = P(B) - P(A)$

- Conditional probability $P(A/B) = \dfrac{P(A \cap B)}{P(B)}, if\ P(B) > 0$

- A and B are independent events if $P(A \cap B) = P(A) \times P(B)$

- Multiplication theorem of probability $P(A \cap B) = \begin{cases} P(A/B) \times P(B), if\ P(B) > 0 \\ P(B/A) \times P(A), if\ P(A) > 0 \end{cases}$

- Bayes' theorem $P(B_j/A) = \dfrac{P(A/B_j) \times P(B_j)}{\sum_{i=1}^{n} P(A/B_j) \times P(B_j)}, j = 1, 2, \ldots n$